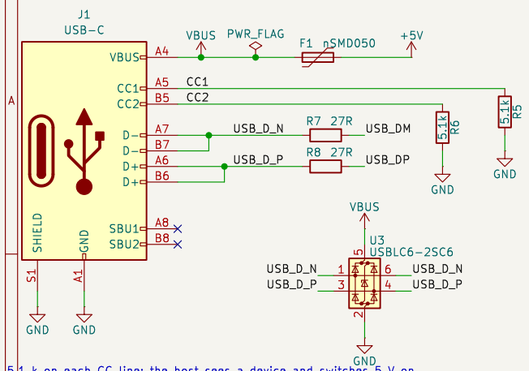
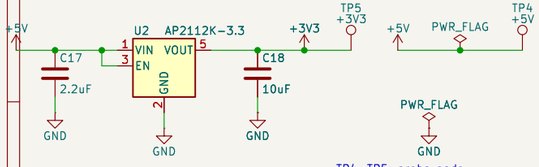
USB-C input, ESD protection, fuse



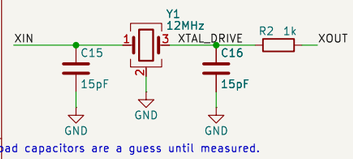
5.1 k on each CC line: the host sees a device and switches 5 V on.

3.3 V regulator



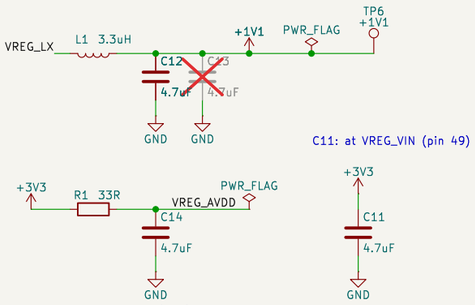
TP4, TP5: probe pads.

12 MHz crystal



Load capacitors are a guess until measured.

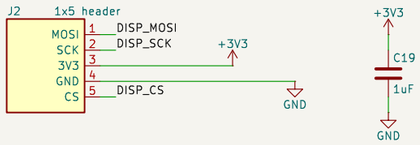
1.1 V core supply: the switching regulator inside U1



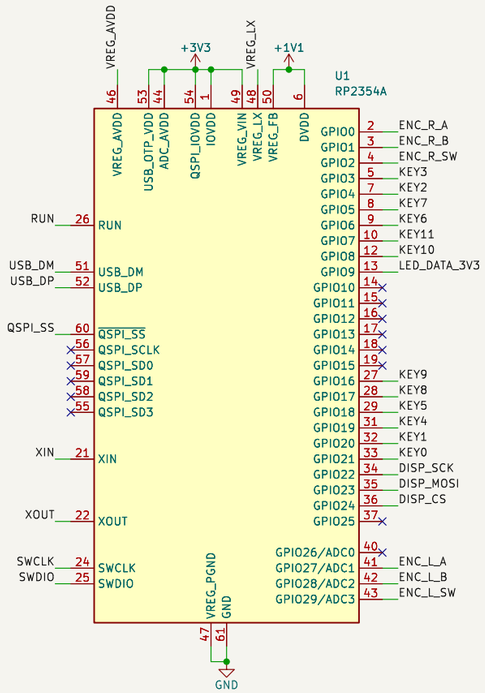
C13: do not populate. COUT (C12) is the one 4.7 uF.

L1: dot on pad 1, the +1V1 end. Not verified on a part.

Display socket for the nice!view module

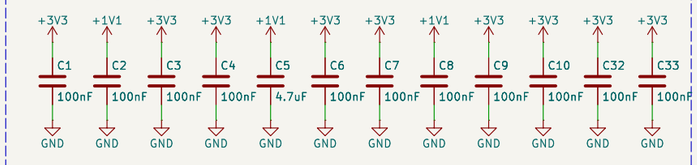


MCU: RP2354A (RP2350 with 2 MB flash inside)

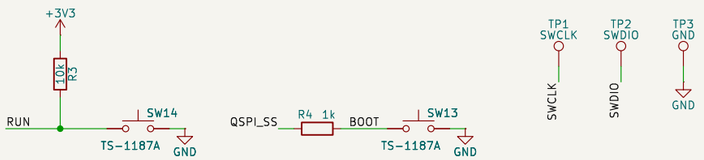


QSPI_SCLK, QSPI_SD0–3: the flash is inside the package. Leave open.

MCU decoupling: 100 nF for each supply pin, 4.7 uF at pin 23



Reset, boot select, debug pads

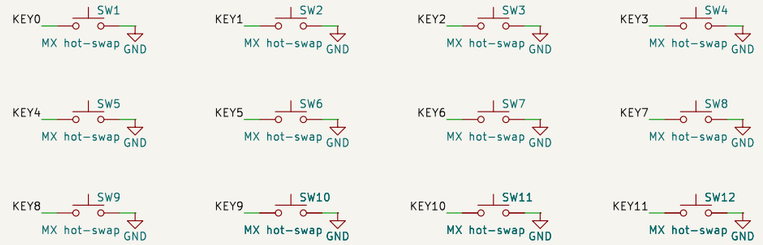


Hold BOOT while plugging in: USB drive for the UF2 file.

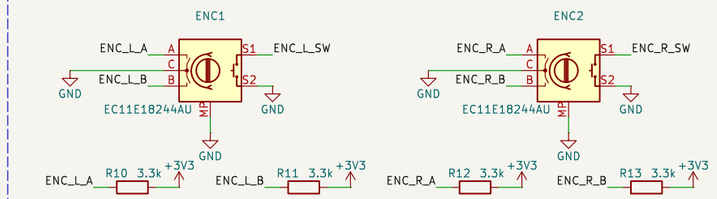
Mounting holes (M2), case ground at H1, fiducials



Keys: one GPIO each, switch to ground, pull-ups inside U1

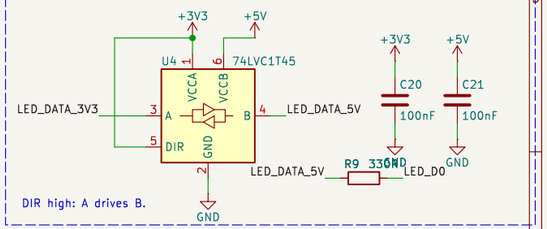


Rotary encoders with push switch, 3.3 k pull-ups on A and B



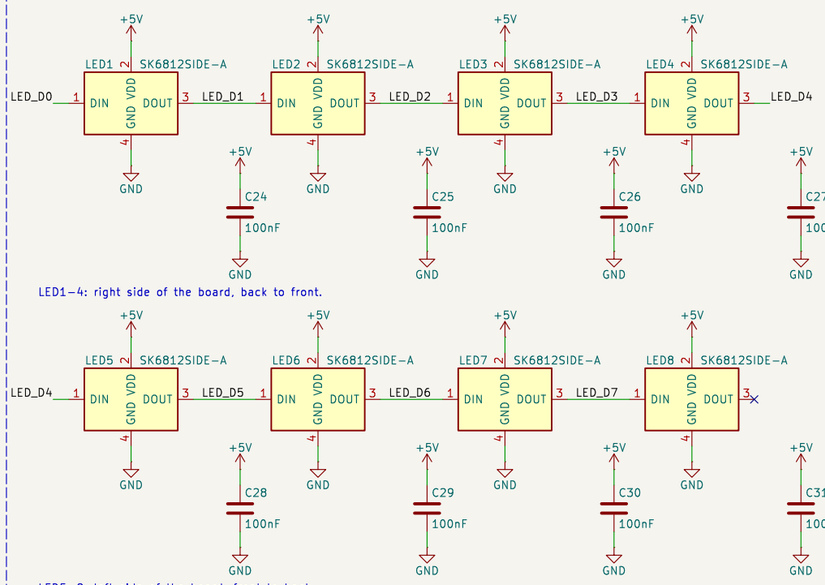
Pull-ups: 1 mA through each contact, the minimum Alps gives. SW keeps U1's own pull-up.

LED data: 3.3 V to 5 V



DIR high: A drives B.

LED chain: 8 side-emitting SK6812, one 100 nF each



LED1–4: right side of the board, back to front.

LED5–8: left side of the board, front to back.

Not built. Nothing on this sheet has been measured on hardware.
Generated from pcb/gen/circuit.py. Do not edit: change circuit.py and run make pcb-place.

worxbend

Sheet: /
File: macropad.kicad_sch

Title: macropad

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 1/1